

Challenge - Build a Code Executor Sandbox

The Problem

Build an HTTP API that executes arbitrary Python code submitted by users. The code should run asynchronously with real-time streaming output.

Requirements

API Design:

python

```
Python
POST /execute
{
  "code": "print('hello world')"
}
→ {"task_id": "abc123"}

GET /tasks/{task_id}/stream # Stream stdout as task runs
GET /tasks/{task_id}       # Get final result + status
```

Your executor will run **untrusted code from the internet**. Users may submit code that is buggy, inefficient, or intentionally malicious.

Your job is to make the system as safe and robust as possible within 2 hours.

What we'd like to see:

- **Correctness:** Does it execute valid code properly?
- **Robustness:** How does it handle problematic code?
- **Security:** What protections exist against malicious code?
- **Design:** Code quality, architecture decisions, trade-offs

Submission

Please provide:

1. **Working code** with setup instructions (README)
2. **Your approach** and any notes you want to share

3. **Test examples** - Show us 2-3 examples of problematic code you tested against and how your system handles them

Note: We don't expect a production-grade sandbox in 2 hours. We want to see your thought process, priorities, and how you approach a hard problem with limited time.

You should treat this as any task you would at work and feel free to use any tools, libraries you may use on a day-to-day basis.

Good luck! We're excited to see your submission.